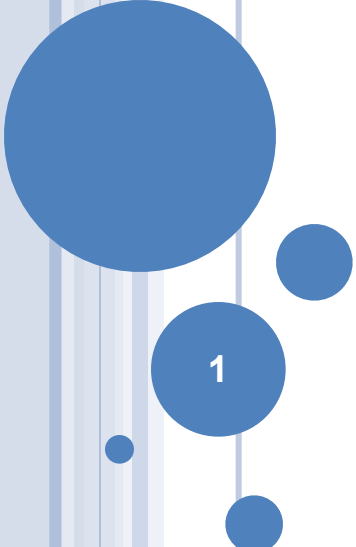


https://hackmd.io/6_KUxQtRQr6H48S7U5OL1Q?view

INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION



Chapter 9 Large-Sample Tests of Hypotheses

https://colab.research.google.com/drive/1TrMHmr5xo5a9-9y6O5t_NM6_zpu9ODyv#scrollTo=kA_ZQISj2nw-

BRAIN-STORMING

- Suppose you draw a fair coin for 100 times.
- Do you think it is common to observe 10 heads?
- Do you think it is common to observe 55 heads?
- Why or why not?

KEY INGREDIENTS

- Model specification: $\mathcal{F}(\theta)$, and want to know θ from your scenario.
- Find an **estimator** of θ
- Standardize the estimator to obtain a **statistic**, and identify its sampling distribution
- Then, we can do two statistical approaches
 - Confidence interval (Ch 8)
 - Hypothesis test (Ch 9 and after)

A decorative graphic on the left side of the slide. It features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle is particularly large and prominent.

9.1.1 HYPOTHESIS TESTS

假設檢定

4

H_0 VS H_a

- H_0 , null hypothesis 虛無假設
 - A contradiction of the alternative hypothesis
 - $H_0 : \theta = \theta_0$
- H_a (or H_1), Alternative hypothesis 對立假設
 - the hypothesis the researcher wants to support
 - $H_1 : \theta \neq \theta_0$

ACTIONS

Only two actions:

1. **Reject $H_0 \neq$ accept H_a ,**
Reject $H_0 \approx$ accept H_a
2. **Do not reject $H_0 \neq$ accept H_0**
Do not reject $H_0 \approx$ accept H_0

TYPE I AND TYPE II ERRORS

- **Type I error**
refers to
reject H_0 when
it is in fact
true.
- **Type II error**
refers to
accept H_0
when it is in
fact false.

	H_0 true	H_0 False (H_a true)
Reject H_0	Type I error	Correct action
Do not reject H_0	Correct action	Type II error

- The significance level 顯著水準 α is the probability of rejecting H_0 when it is in fact true.
 - Or, the probability of Type I error:
$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) = P(\text{Type I error})$$
- β is the probability of accepting H_0 when it is in fact false:
$$\beta = P(\text{accept } H_0 \mid H_0 \text{ false}) = P(\text{Type II error})$$
- The power of the test is $(1 - \beta)$, the probability of rejecting H_0 when it is false.

STANDARD OPERATION PROCEDURES

1. Set up H_0 vs. H_a .
2. Decide the significant level α (usually $\alpha=0.05, 0.1, 0.01$).
3. Find a test statistic and its sampling distribution.
4. Calculate the realized test statistic from the data.
5. Find one between
 1. the rejection region (RR)
 2. the p -value.
6. Conclude

The Rejection region guarantees that
 $P(\text{Type I error}) = \alpha$

A TEST STATISTIC

1. A_test statistic

1. 檢定統計量,

2. A standardized estimator

3.
$$Z(\hat{\theta}) = \frac{\hat{\theta} - \theta_0}{SE}$$

2. Its sampling distribution is $N(0,1)$ by CLT.

$$\left(Z(\hat{\theta}) = \frac{\hat{\theta} - \theta_0}{SE} \right) \xrightarrow{d} Z$$

(A) THE REJECTION REGION APPROACH (OR CRITICAL VALUE APPROACH)

5. Find the rejection region:

$$\mathbb{P} \left(Z(\hat{\theta}) \in \text{RR} \mid H_0 \text{ is true} \right) = \alpha$$

6. Conclusion:

- If the realized statistics falls into the rejection region, reject H_0 .
- Otherwise, do not reject H_0 .

(B) THE P-VALUE APPROACH

5. The p -value is the probability of observing a test statistic as extreme as or more than the one observed assuming H_0 is true.

p -value

= $\mathbb{P}(\text{Observing a realized test statistic or more extreme values} \mid H_0 \text{ true})$

6. Conclusion:

- If p -value is smaller than α , reject H_0 .
- Otherwise, do not reject H_0

A decorative vertical bar on the left side of the slide, composed of several thin, parallel stripes in various shades of blue. To the right of this bar, there are several overlapping circles of different sizes, also in shades of blue, creating a modern, abstract design.

9.2 ONE POPULATION MEAN

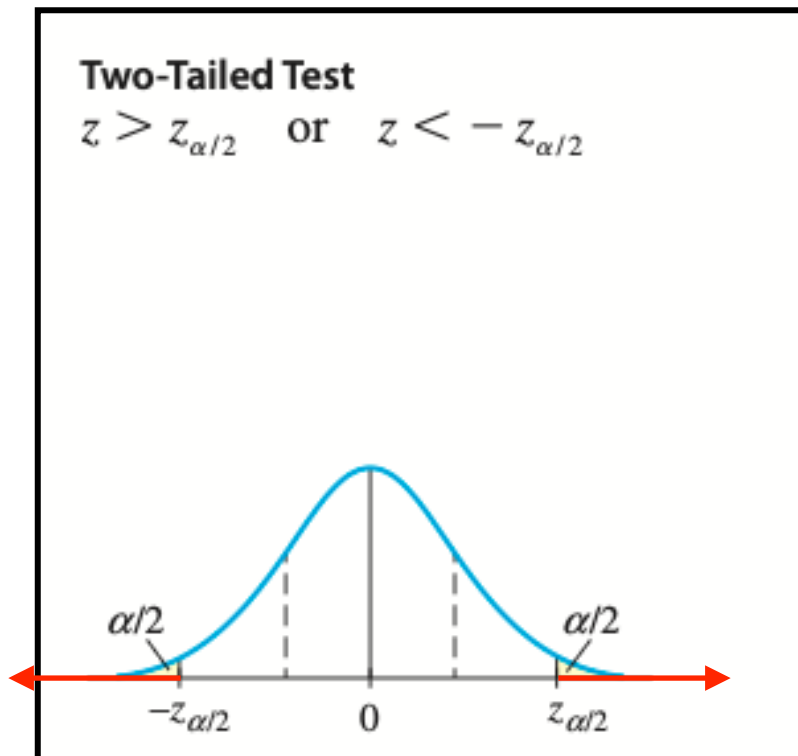
13

The RR approach guarantees that $P(\text{Type I error}) = \alpha$

TWO-SIDED TEST

- $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$

$$\mathbf{RR} = \{z : z > z_{\alpha/2}, \text{ or } z < -z_{\alpha/2}\} = \{z : |z| > z_{\alpha/2}\}$$



$z_{\alpha/2}$ is called the *critical value*.
Thus, RR approach is also called critical value approach.

The RR approach guarantees that $P(\text{Type I error}) = \alpha$

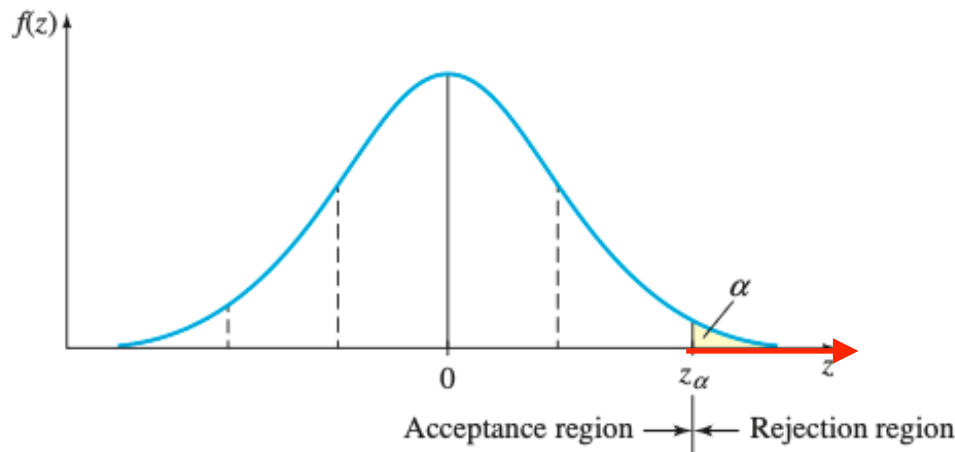
RIGHT-SIDED TEST

- $H_0 : \mu = \mu_0$ versus $H_a : \mu > \mu_0$
- $H_0 : \mu \leq \mu_0$ versus $H_a : \mu > \mu_0$
- $H_0 : \mu < \mu_0$ versus $H_a : \mu \geq \mu_0$

RR = $\{z : z > z_\alpha\}$ z_α is called the *critical value*. Thus, RR approach is also called critical value approach.

Figure 9.3

The rejection region for a right-tailed test with significance level α



The RR approach guarantees that $P(\text{Type I error}) = \alpha$

LEFT-SIDED TEST

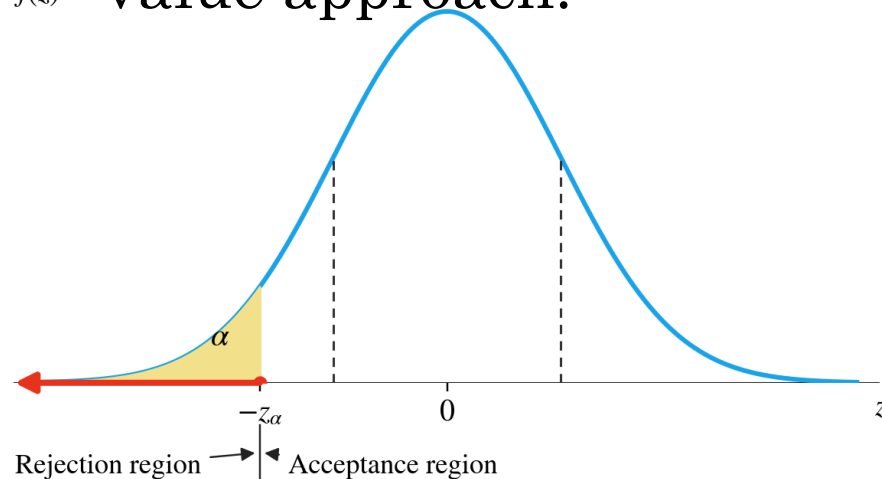
- $H_0 : \mu = \mu_0$ versus $H_a : \mu < \mu_0$
- $H_0 : \mu \geq \mu_0$ versus $H_a : \mu < \mu_0$
- $H_0 : \mu > \mu_0$ versus $H_a : \mu \leq \mu_0$

RR = $\{z : z < -z_\alpha\}$ z_α is called the *critical value*. Thus, RR approach is also called critical

value approach.

Left-tailed test

The rejection region for a left-tailed test with significance level α



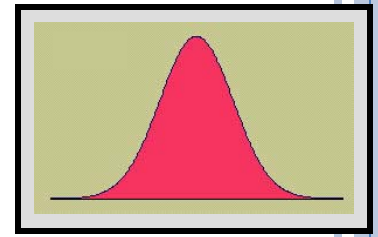
P-VALUE APPROACH

1. RR approach needs to check Tables at a given α
2. When computation is simple, p -value approach is preferred

人工時期與機械時期計算工具之演進表

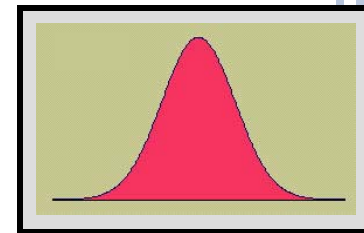
	工具名稱	年代(西元)	發明者	說明
人工時期	算盤	約1100年	中國人	人類最早的計算工具
	計算尺	1620	艾德蒙·甘特 (Edmund Gunter)	西方早期常用的計算工具
機械時期	加法器	1642	巴斯卡 (Blaise Pascal)	以齒輪帶動計數轉盤，大小如香煙盒
	自動進位加法器	1673	萊布尼茲 (Gottfried Leibniz)	可連續加、乘
	四則運算機	1820	湯瑪斯 (C.X. Thomas)	可做加、減、乘、除運算
	差分機	1833	巴貝基 (Charles Babbage)	具有現代電腦輸入、輸出處理功能，由於巴貝基對於計算工具貢獻良多，因此被尊稱為計算機之父
	打孔卡片	1890	何樂禮 (Herman Hollerith)	以打孔位置來儲存資料
	馬克一號計算機 (MARK-1)	1944	艾肯 (Howard Aiken)	史上最大的電力機械計算機 以電磁繼電器做成

WE PREFER TO REPORT P-VALUE!



- The p -value approach is often preferred because
 - You can evaluate the test results at any significance level you choose.
- What should you do if you are the experimenter and no one gives you a significance level to use?

STATISTICAL SIGNIFICANCE



- If the p -value is less than **.01**, reject H_0 . The results are **highly significant**.
- If the p -value is between **.01** and **.05**, reject H_0 . The results are **statistically significant**.
- If the p -value is between **.05** and **.10**, do not reject H_0 . But, the results are **tending towards significance**.
- If the p -value is greater than **.10**, do not reject H_0 . The results are **not statistically significant**.

AN EXAMPLE

Table 5.2: **Static features of selected stocks and exchange rate.** This table provides the static features with a study period from 2008/1/2 to 2021/12/31, including the first four moments, minimum, maximum and the Jarque-Bera statistic.

Stock	mean	sd	skew	kurt	min	max	JB
6505	0.00014	0.0167	0.04	3.26***	-0.1056	0.0945	1518***
2201	0.00003	0.0215	0.34***	3.64***	-0.1047	0.1199	1965***
1303	0.00019	0.0161	-0.14***	3.26***	-0.1024	0.0745	1533***
1301	0.00025	0.0154	-0.04	3.41***	-0.0971	0.0688	1663***

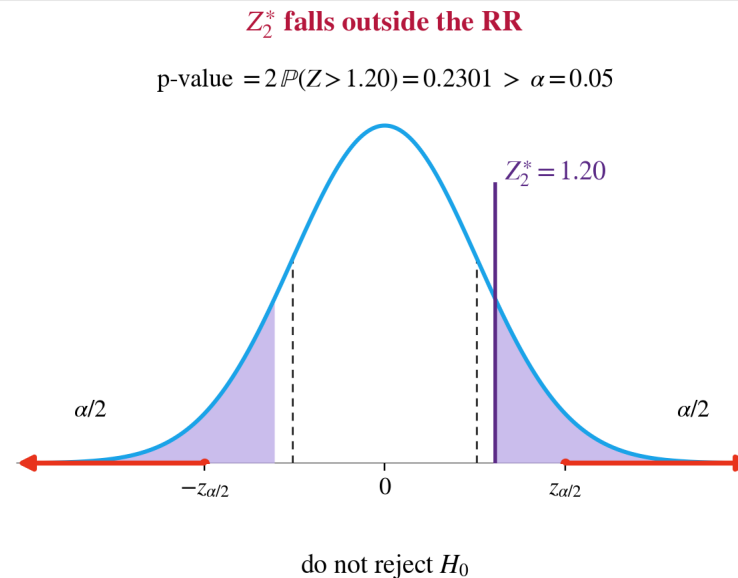
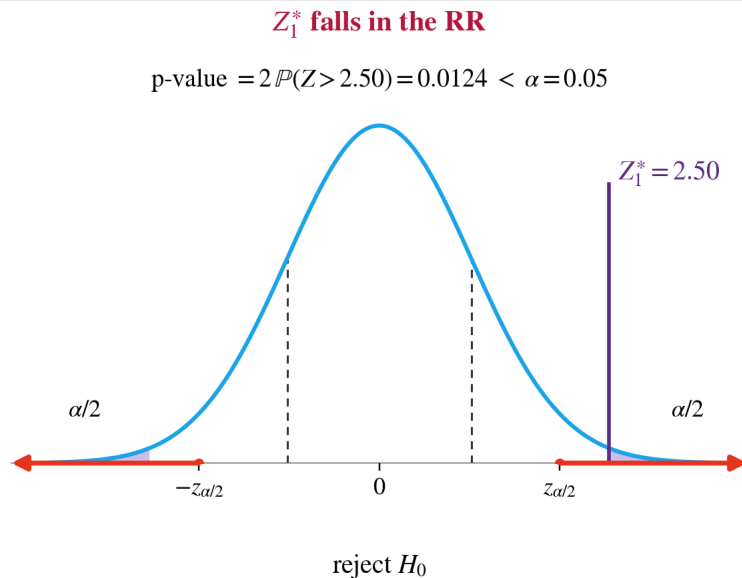
Tables 5.2 and 5.3 provide static and dynamic measures for the selected stocks and the exchange rate, respectively. Here, *, ** and *** denote the significant **level** 10%, 5% and 1%, respectively. Table 5.2 shows the static features, including the first four moments, minimum return, maximum return and the Jarque-Bera statistic. Among them, the mean, skewness and kurtosis are tested for the **significance**. The Jarque-Bera tests whether the given data follows a normal distribution. The results show that all stock returns do not follow a normal distribution under 1% significant **level**.

TWO-SIDED TEST

- $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$

$$\text{RR} = \{z : z > z_{\alpha/2}, \text{ or } z < -z_{\alpha/2}\} = \{z : |z| > z_{\alpha/2}\}$$

$$p\text{-value} = \mathbb{P}(Z > |Z^*|) + \mathbb{P}(Z < -|Z^*|) = 2 \times \mathbb{P}(Z > |Z^*|)$$



RIGHT-SIDED TEST

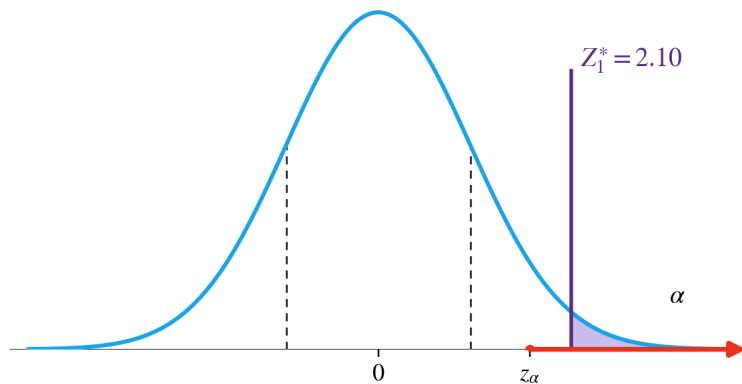
- $H_0 : \mu = \mu_0$ versus $H_a : \mu > \mu_0$
- $H_0 : \mu \leq \mu_0$ versus $H_a : \mu > \mu_0$
- $H_0 : \mu < \mu_0$ versus $H_a : \mu \geq \mu_0$

$$\mathbf{RR} = \{z : z > z_\alpha\}$$

$$p - \text{value} = \mathbb{P}(Z > Z^*)$$

Z_1^* falls in the RR

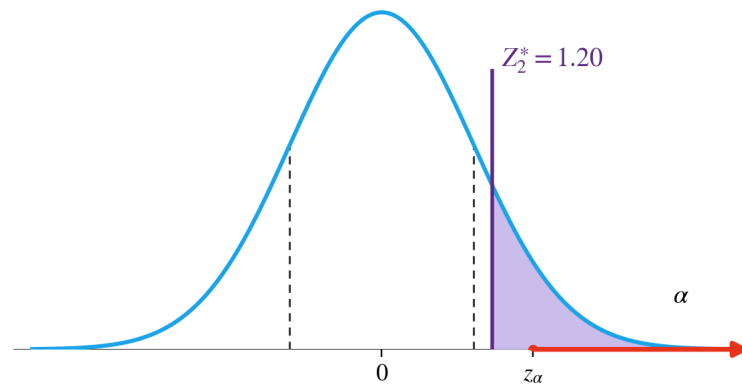
$$\text{p-value} = \mathbb{P}(Z > 2.10) = 0.0179 < \alpha = 0.05$$



reject H_0

Z_2^* falls outside the RR

$$\text{p-value} = \mathbb{P}(Z > 1.20) = 0.1151 > \alpha = 0.05$$



do not reject H_0

If Z^* falls in the RR $\iff$ p-value $< \alpha$

LEFT-SIDED TEST

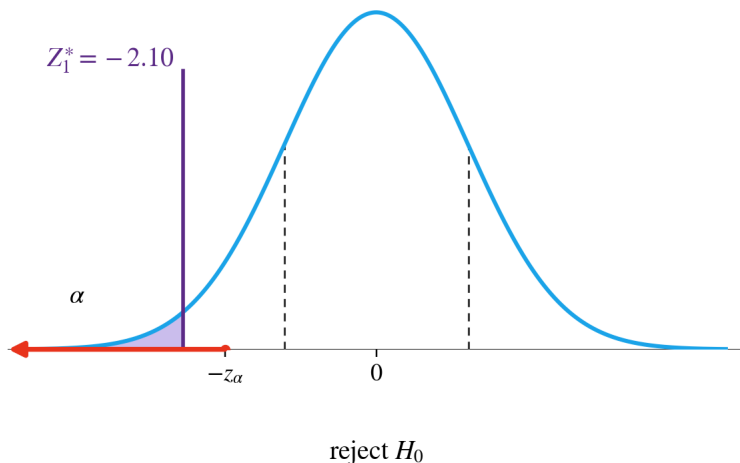
- $H_0 : \mu = \mu_0$ versus $H_a : \mu < \mu_0$
- $H_0 : \mu \geq \mu_0$ versus $H_a : \mu < \mu_0$
- $H_0 : \mu > \mu_0$ versus $H_a : \mu \leq \mu_0$

$$\text{RR} = \{z : z < -z_\alpha\}$$

$$p\text{-value} = \mathbb{P}(Z < Z^*)$$

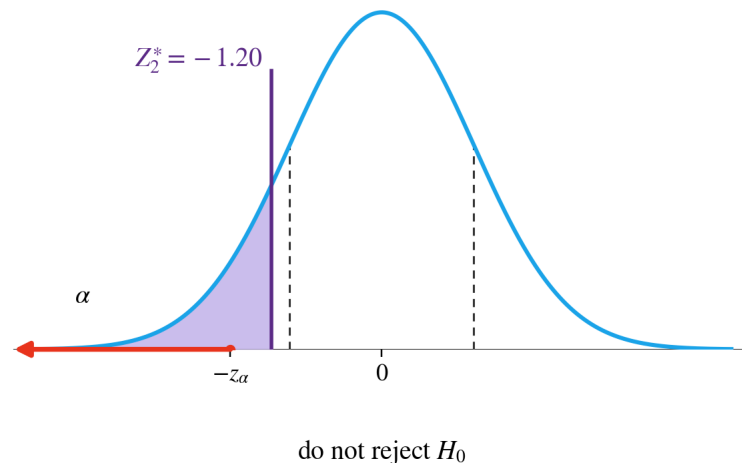
Z_1^* falls in the RR

$$p\text{-value} = \mathbb{P}(Z < -2.10) = 0.0179 < \alpha = 0.05$$



Z_2^* falls outside the RR

$$p\text{-value} = \mathbb{P}(Z < -1.20) = 0.1151 > \alpha = 0.05$$



KEY INGREDIENTS (A1)

- Model: $X_i \stackrel{i.i.d.}{\sim} \mathcal{F}(\mu, \sigma^2)$ for $i = 1, \dots, n$.
- Our interest: μ .
- Use $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ to estimate μ .

KEY INGREDIENTS (A2)

- CLT:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} Z$$

- When σ is unknown, we use the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

The Slutsky's theorem:

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \xrightarrow{d} Z$$

$$H_0 : \mu = \mu_0$$

- The test statistics and its sampling distribution is

$$Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \xrightarrow{d} Z.$$

THE TWO-SIDED TEST

1. $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$
2. Set up α
3. $Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \xrightarrow{d} Z$
4. Calculate the realized statistic Z^* .
5. Find (a) or (b)
 - (a) The rejection region = $\{z : z < -z_{\alpha/2}, z > z_{\alpha/2}\}$
 - (b) The p -value = $2P(Z > |Z^*|)$.
6. Conclude.
 - (a) If Z^* falls into the rejection region, we reject H_0 .
 - (b) If $p\text{-value} < \alpha$, we reject H_0 .

THE LEFT-SIDED TEST

1. One of the following:

- $H_0 : \mu = \mu_0$ vs $H_a : \mu < \mu_0$
- $H_0 : \mu \geq \mu_0$ vs $H_a : \mu < \mu_0$
- $H_0 : \mu > \mu_0$ vs $H_a : \mu \leq \mu_0$

2. Set up α

$$3. Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \xrightarrow{d} Z$$

4. Calculate the realized statistic Z^* .

5. Find (a) or (b)

(a) The rejection region = $\{z : z < -z_\alpha\}$

(b) The p -value = $P(Z < Z^*)$.

6. Conclude.

THE RIGHT-SIDED TEST

1. One of the following:

- $H_0 : \mu = \mu_0$ vs $H_a : \mu > \mu_0$
- $H_0 : \mu \leq \mu_0$ vs $H_a : \mu > \mu_0$
- $H_0 : \mu < \mu_0$ vs $H_a : \mu \geq \mu_0$

2. Set up α

$$3. Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \xrightarrow{d} Z$$

4. Calculate the realized statistic Z^* .

5. Find (a) or (b)

(a) The rejection region = $\{z : z > z_\alpha\}$

(b) The p -value = $P(Z > Z^*)$.

6. Conclude.



EXAMPLE

- The daily yield for a chemical plant has averaged 880 tons for several years. The quality control manager wants to know if this average has changed. She randomly selects 50 days and records an average yield of 871 tons with a standard deviation of 21 tons.

$$H_0 : \mu = 880$$

$$H_a : \mu \neq 880$$

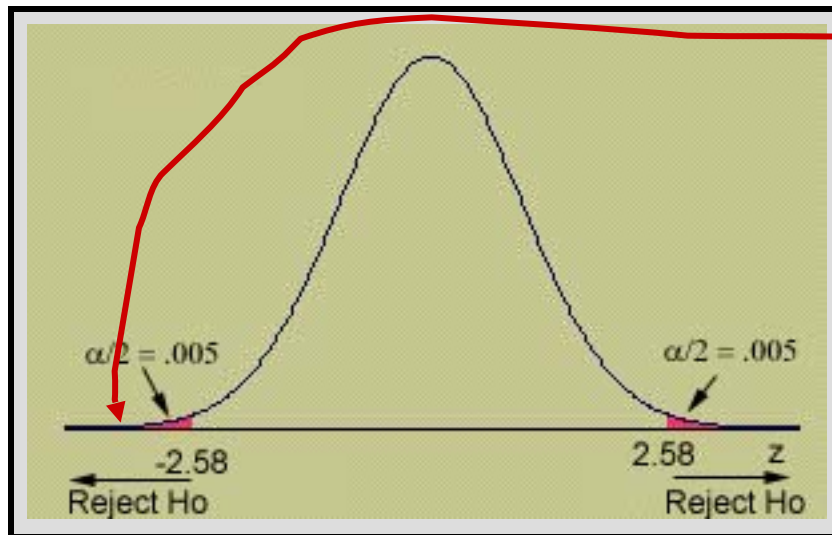
Test statistic :

$$z \approx \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{871 - 880}{21 / \sqrt{50}} = -3.03$$



EXAMPLE

What is the critical value of z that cuts off exactly $\alpha/2 = .01/2 = .005$ in the tail of the z distribution?



For our example, $z = -3.03$ falls in the rejection region and H_0 is rejected at the 1% significance level.

Rejection Region: Reject H_0 if $z > 2.58$ or $z < -2.58$. If the test statistic falls in the rejection region, its p -value will be less than $\alpha = .01$.

EXAMPLE



Choose a significance level, say $\alpha = .01$

5. The rejection region is

$$\{z : z < -2.58 \text{ or } z > 2.58\}$$

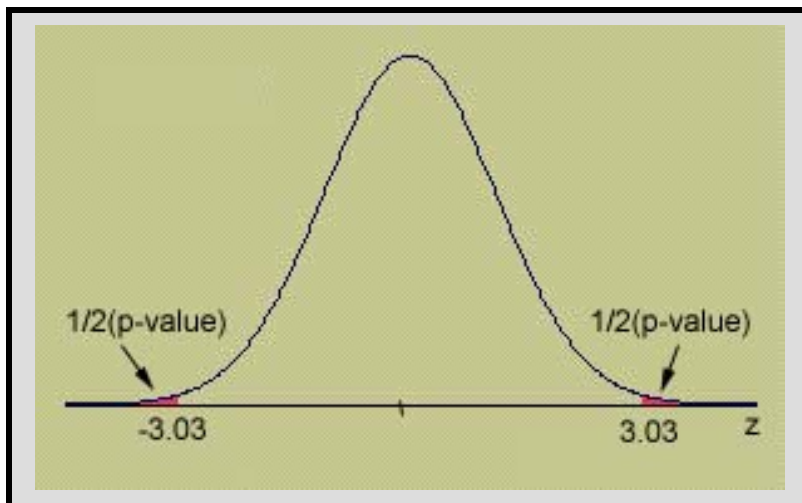
6. Because $z^* = -3.03$ falls into the rejection region, we reject H_0



EXAMPLE

What is the probability that this test statistic or something even more extreme (far from what is expected if H_0 is true) could have happened *just by chance*?

$$p\text{-value: } P(z > 3.03) + P(z < -3.03) \\ = 2P(z < -3.03) = 2(.0012) = .0024$$



This is an unlikely occurrence, which happens about 2 times in 1000, assuming $\mu = 880$!

EXAMPLE



- To make our decision clear, we choose a significance level, say $\alpha = .01$.

If the p -value is less than α , H_0 is rejected as false. You report that the results are statistically significant at level α .

If the p -value is greater than α , H_0 is not rejected. You report that the results are not significant at level α .

Since our p -value $= .0024$ is less than α , we reject H_0 and conclude that the average yield has changed.



EXAMPLE

- A homeowner randomly samples 64 homes similar to her own and finds that the average selling price is \$252,000 with a standard deviation of \$15,000. Is this sufficient evidence to conclude that the average selling price is greater than \$250,000? Use $\alpha = .01$.
- Sol.  Reiterate the problem:
 - $X_i \sim \mathcal{G}(\mu, \sigma^2)$ for $i = 1, \dots, 64$. $\mathcal{G}$ is unknown.
 - $\bar{x} = 252,000$, $s = 15,000$.

1. $H_0 : \mu = 250,000$ vs $H_1 : \mu > 250,000$

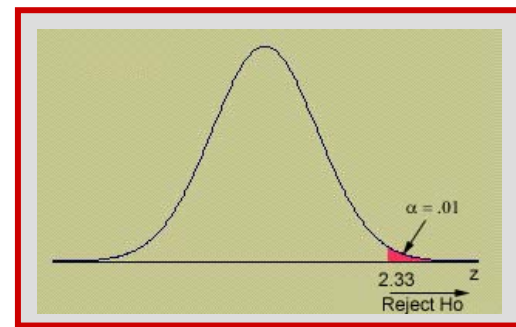
2. Set $\alpha = 0.01$

3. The test statistic is $Z_{STAT} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \rightarrow Z$.

$$4. Z^* = \frac{252000 - 250000}{15000/\sqrt{64}} = 1.07$$

5. Rejection region = $\{z : z > 2.33\}$

6. Because Z^* does not fall into RR. Do not reject H_0 .



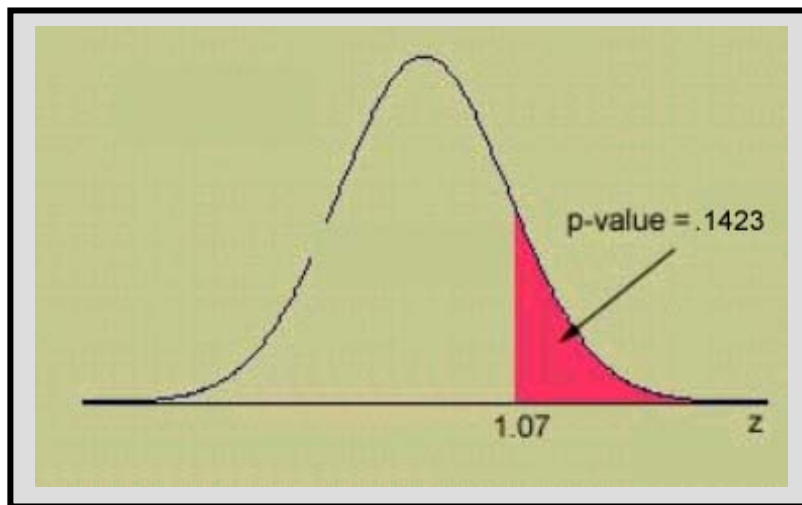
正面表述：Average price is likely about 250,000.

負面表述：There is not enough evidence to indicate that μ is greater than \$250,000.

P-VALUE APPROACH



$$p\text{-value: } P(z > 1.07) = 1 - .8577 = .1423$$



Since the p -value is greater than $\alpha = .01$, H_0 is not rejected.

There is insufficient evidence to indicate that μ is greater than \$250,000.

TWO TYPES OF ERRORS



We want to keep the probabilities of error as small as possible.

- **The value of α is the significance level, and is controlled by the experimenter.**
- **The value of β is difficult, if not impossible to calculate.**

Rather than “accepting H_0 ” as true without being able to provide a measure of goodness, we choose to “not reject” H_0 .

We write: There is insufficient evidence to reject H_0 .



9.2.3 POWER OF A TEST (SKIPPED, SEE YOUTUBE VEDIOS)

A decorative graphic on the left side of the slide consists of several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle contains the number 40.

9.3 TWO POPULATION MEANS (SKIPPED 20251122)

40

A decorative vertical bar on the left side of the slide, composed of several thin, parallel vertical stripes in various shades of blue. To the right of this bar, there are several blue circles of different sizes. One large circle is positioned near the top, and several smaller circles are arranged vertically below it, some overlapping the stripes.

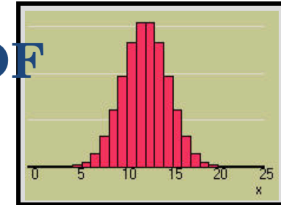
9.4 POPULATION PROPORTION

41

RECALL

Chapter 7.5

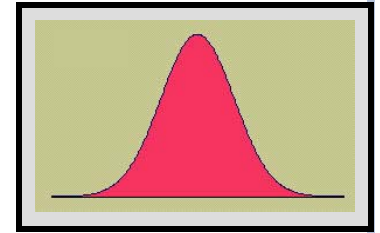
THE SAMPLING DISTRIBUTION OF THE SAMPLE PROPORTION



- Assume $X_i \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$
- If n is large, and p is not too close to zero or one,
By CLT,

$$\frac{\hat{p} - p}{\sqrt{pq/n}} \xrightarrow{d} N(0,1)$$

TESTING A BINOMIAL PROPORTION P



A random sample of size n from a binomial population to test

$H_0 : p = p_0$ versus

H_a : one of three alternatives

$$\text{Test statistic : } z \approx \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$$

with rejection regions and/or p - values based on the standard normal z distribution.



EXAMPLE

- Regardless of age, about 20% of American adults participate in fitness activities at least twice a week. A random sample of 100 adults over 40 years old found 15 who exercised at least twice a week. Is this evidence of a decline in participation after age 40? Use $\alpha = .05$.

$$H_0 : p = .2$$

$$H_a : p < .2$$

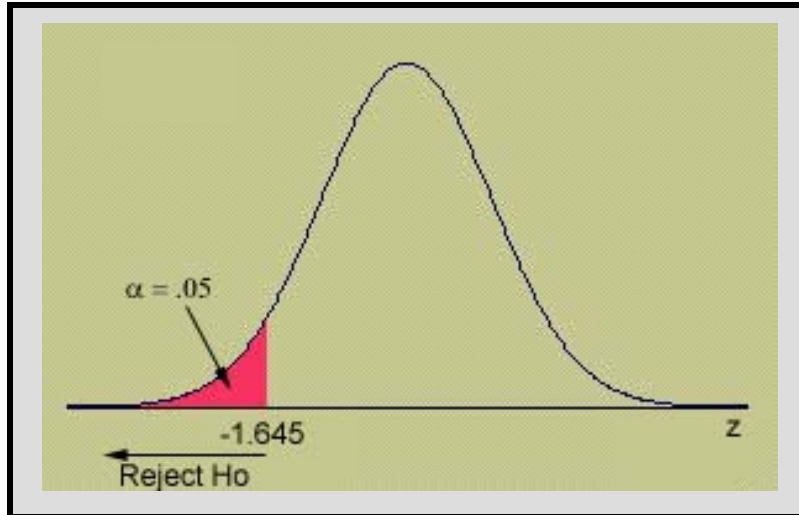
Test statistic :

$$z \approx \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{.15 - .2}{\sqrt{\frac{.2(.8)}{100}}} = -1.25$$

CRITICAL VALUE APPROACH



What is the critical value of z that cuts off exactly $\alpha = .05$ in the left-tail of the z distribution?



For our example, $z = -1.25$ does not fall in the rejection region and H_0 is not rejected. There is not enough evidence to indicate that p is less than .2 for people over 40.

Rejection Region: Reject H_0 if $z < -1.645$. If the test statistic falls in the rejection region, its p -value will be less than $\alpha = .05$.

EXAMPLE 1

- Suppose you draw a coin for 100 times and observe 10 heads.
- Set $\alpha = 0.05$.
- Set $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$.

Sol. 

(a) Calculate

p-value

$= P(\text{Observe 10 heads or more extreme values} \mid H_0 \text{ is true})$

$$= 2 \times \mathbb{P}(Z > |\frac{0.1 - 0.5}{0.05}|) = 2 \times P(Z > 8) = 3.06 \times 10^{-17}$$

(b) Do you reject H_0 or not? **Yes. I reject H_0 .**

(c) Please conclude. **p is likely to be not equal to 0.5.**

EXAMPLE 2-1

- Suppose you draw a coin for 100 times and observe 40 heads.
- Set $\alpha = 0.05$.
- Set $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$.

Sol. 

(a) Calculate *p-value*

$= P(\text{Observe 40 heads or more extreme values} \mid H_0 \text{ is true})$

$$= 2 \times \mathbb{P} \left(Z > \left| \frac{0.4 - 0.5}{\sqrt{0.5^2/100}} \right| \right) = 2 \times \mathbb{P}(X > 2) = 0.0456$$

(b) Do you reject H_0 or not? Yes. I reject H_0 .

(c) Please conclude. p is likely to be not 0.5.

EXAMPLE 2-2

- Suppose you draw a coin for 100 times and observe 40 heads.
- Set $\alpha = 0.05$.
- Set $H_0 : p = 0.5$ vs $H_1 : p > 0.5$.

Sol. 

Calculate *p-value*

$= P(\text{Observe 40 heads or more extreme values} \mid H_0 \text{ is true})$

$$= \mathbb{P} \left(Z > \frac{0.4 - 0.5}{\sqrt{0.5^2/100}} \right) = \mathbb{P}(X > -2) = \mathbf{0.9772}$$

(a) Do you reject H_0 or not? **No. I don't reject H_0**

(b) Please conclude. **p is likely to be 0.5.**

EXAMPLE 2-3

- Suppose you draw a coin for 100 times and observe 40 heads.
- Set $\alpha = 0.05$.
- Set $H_0 : p = 0.5$ vs $H_1 : p < 0.5$.

Sol. 

Calculate *p-value*

$= P(\text{Observe 40 heads or more extreme values} \mid H_0 \text{ is true})$

$$= \mathbb{P} \left(Z < \frac{0.4 - 0.5}{\sqrt{0.5^2/100}} \right) = \mathbb{P}(X < -2) = 0.0228$$

(a) Do you reject H_0 or not? Yes, I reject H_0 .

(b) Please conclude. p is likely to be less than 0.5.

The left side of the slide features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several blue circles of different sizes. One large circle is positioned near the top, and several smaller circles are arranged vertically below it, some overlapping the stripes.

9.5 TWO POPULATION PROPORTIONS (SKIPPED 20251122)

50